~~**G**eneralised~~

Linear

~~**M**ixed-Effects~~

Models

What is a Linear Model?

A linear model describes the relationship between a **continuous dependent variable** (response) and one or more **independent variables** (predictors) using a linear equation:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i$$

- y_i – outcome for observation i
- x_{ij} – predictor variables
- β_j – regression coefficients (estimated from data)
- ε_i – random noise (Gaussian-distributed)
  (error $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$)

- Predicting a **continuous** outcome from **independent** predictors
- E.g. Predicting mouse weight from genotype and diet

- Estimate the **best-fitting line (or plane)** through the data
- Using **Ordinary Least Squares (OLS)** to minimize prediction error

$$\min \textstyle\sum_i (y_i - \hat{y}_i)^2$$

What Does “Linear” Really Mean?

Not about the shape of the predictors!

A **linear model** is linear **in the parameters** (the β's), not necessarily in the predictors X themselves.

<u>These are all linear models:</u>

- Simple linear

$$y = \beta_0 + \beta_1 x$$

- Polynomial regression

$$y = \beta_0 + \beta_1 x + \beta_2 x^2$$

- Log-transformed predictor

$$y = \beta_0 + \beta_1 \log(x)$$

- Categorical predictors

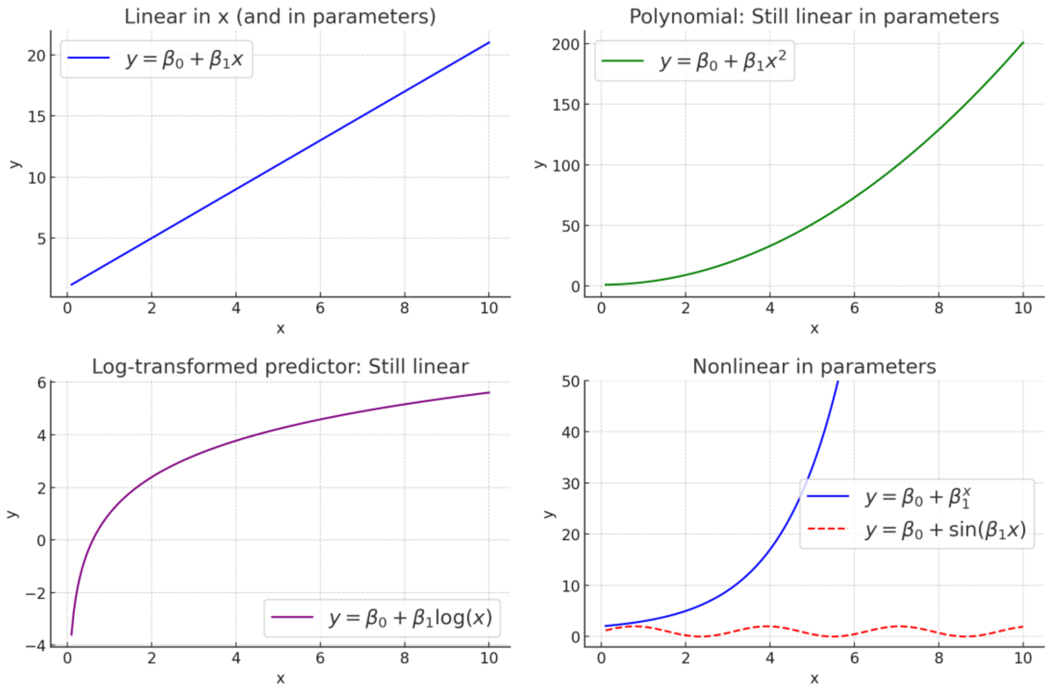
$$y = \beta_0 + \beta_1 \cdot \mathrm{Male}$$

<u>Nonlinear models example:</u>

- $y = \beta_0 + \beta_1^x$ — not linear in parameters
- $y = \beta_0 + \sin(\beta_1 x)$ — also nonlinear in β_1

Rule of thumb

If the equation is a **linear combination** of parameters (each multiplied by some function of xxx),
→ it's a linear model!



Model Formulas – From Math to Code

Math Expression	Formula Syntax (`R` , `statsmodels` , `fitlm`)	Description
$y = \alpha$	`y ~ 1`	Intercept only (null model)
$y = \alpha + \beta_1 x_1$	`y ~ 1 + x1` or `y ~ x1`	Intercept + slope for x_1
$y = \beta_1 x_1$ (no intercept)	`y ~ 0 + x1` or `y ~ -1 + x1`	Slope only, no intercept
$y = \alpha + \beta_1 x_1 + \beta_2 x_2$	`y ~ x1 + x2`	Two additive effects
$y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$	`y ~ x1 * x2`	Main effects + interaction
$y = \alpha + \beta_{12} x_1 x_2$ only	`y ~ x1:x2`	Interaction only
$y = \alpha + \beta_1 x_1 + \beta_{12} x_1 x_2$	`y ~ x1 + x1:x2`	Partial interaction (only x_1)

Interpreting Linear Model Output

Term	Meaning
Estimate ($\hat{\beta}$)	Effect size of predictor on outcome
Std. Error	Uncertainty of estimate (standard deviation)
t-value	Ratio $\hat{\beta}/\mathrm{SE}$, used for testing
p-value	Probability under null hypothesis
R^2	Proportion of variance explained
Residual Std. Error / RMSE	Typical prediction error

Code example

```python
import statsmodels.formula.api as smf
import pandas as pd

# Example data
data = pd.DataFrame({
    'y': [10, 12, 15, 18, 20, 22, 25, 28, 30, 32],
    'x1': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    'x2': [0.5, 1.2, 1.8, 2.3, 2.8, 3.5, 4.0, 4.5, 5.1, 5.7]
})

# Linear model with interaction: y ~ x1 * x2
# This expands to y ~ x1 + x2 + x1:x2
model = smf.ols(formula='y ~ x1 * x2', data=data)

# Fit the model
results = model.fit()

# Print summary
print(results.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.998
Model:                            OLS   Adj. R-squared:                  0.997
Method:                 Least Squares   F-statistic:                     1072.
Date:                Tue, 10 Jun 2025   Prob (F-statistic):           6.38e-09
Time:                        16:48:01   Log-Likelihood:                -13.882
No. Observations:                  10   AIC:                             35.76
Df Residuals:                       6   BIC:                             36.97
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      2.3276      1.734      1.342      0.227      -1.921       6.576
x1             1.5284      0.245      6.241      0.001       0.930       2.127
x2             2.2356      0.547      4.086      0.007       0.902       3.569
x1:x2         -0.1257      0.053     -2.379      0.056      -0.255       0.004
==============================================================================
Omnibus:                        0.329   Durbin-Watson:                   1.937
Prob(Omnibus):                  0.848   Jarque-Bera (JB):                0.407
Skew:                           0.262   Prob(JB):                        0.816
Kurtosis:                       2.368   Cond. No.                         116.
==============================================================================
```

Limitations of Linear Models / Assumptions of Linear Regression

Assume independence of observations

- Cannot handle **repeated measures** or **grouped data**
- Fails when data are **nested** (e.g. trials within subjects)

Ignore between-subject variability

- Force a **single intercept and slope** for all subjects
- Do not account for **individual differences**

No way to model hierarchical structure

- Flat model structure only
- Cannot distinguish within-group vs. between-group effects

Result:
Invalid inference in structured data

- Biased standard errors
- Inflated Type I error
- Loss of generalizability

LM Assumes Independent Observations

LM treats all observations as if they are unrelated.
But real data is often grouped or repeated.

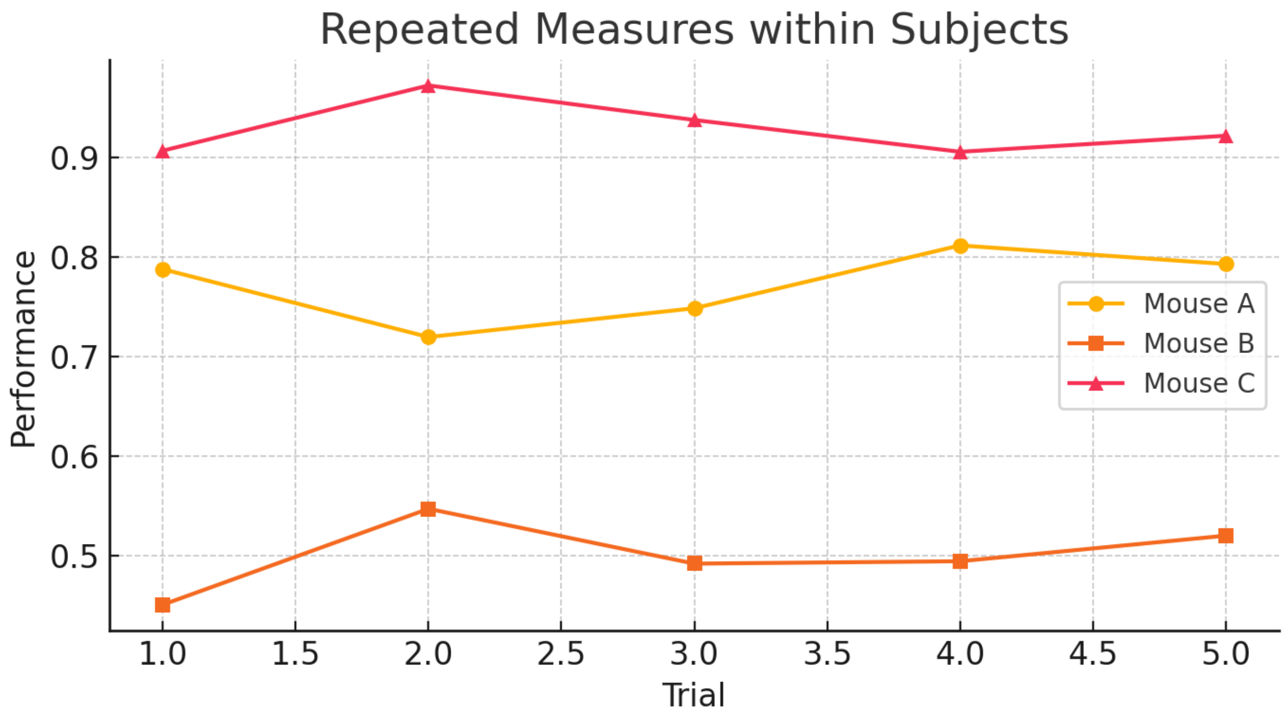
Mouse ID	Trial	Performance
A	1–5	[0.7, 0.8, 0.6, 0.7, 0.8]
B	1–5	[0.5, 0.4, 0.5, 0.5, 0.6]
C	1–5	[0.9, 0.9, 1.0, 0.8, 0.9]

LM assumes 15 independent data points

But trials **within a mouse are correlated**

👉 **Violation** of independence

→ Wrong SE, wrong inference

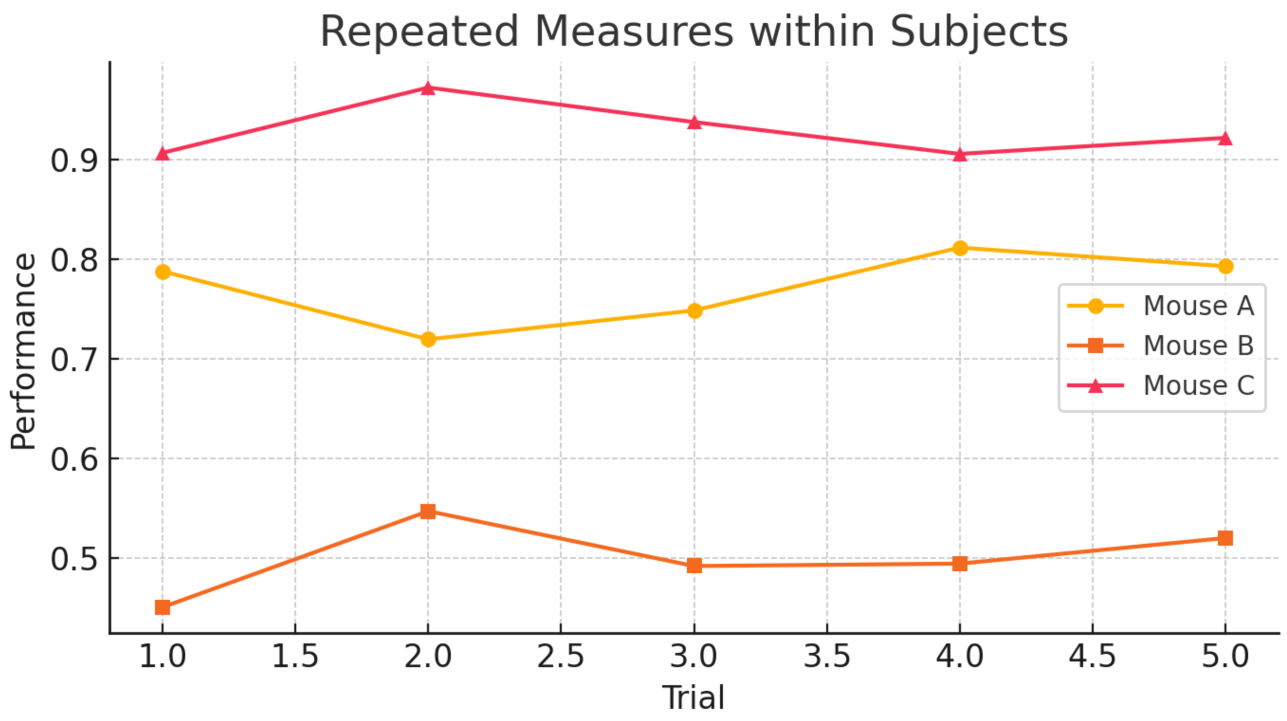


LM Cannot Model Individual Differences:

LM fits one intercept and slope for everyone.

All subjects are assumed to follow the **same trend**

$$\text{Performance} = \beta_0 + \beta_1 \cdot \text{Trial}$$



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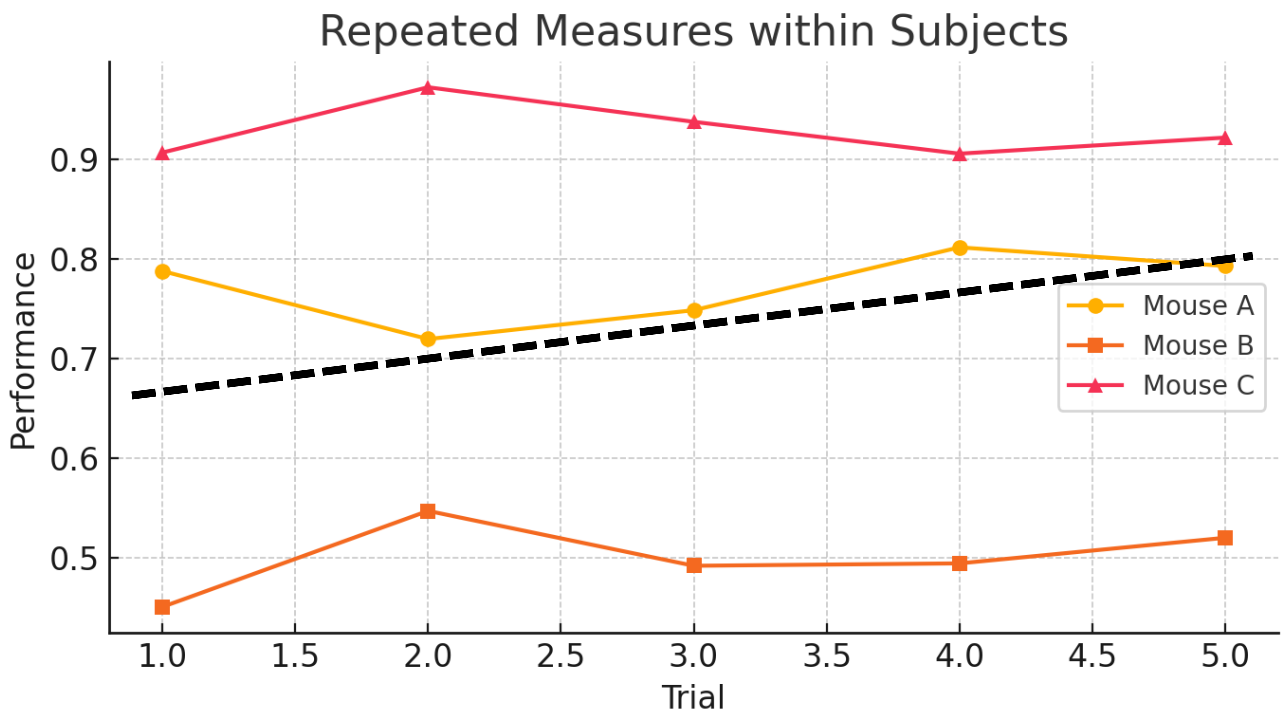

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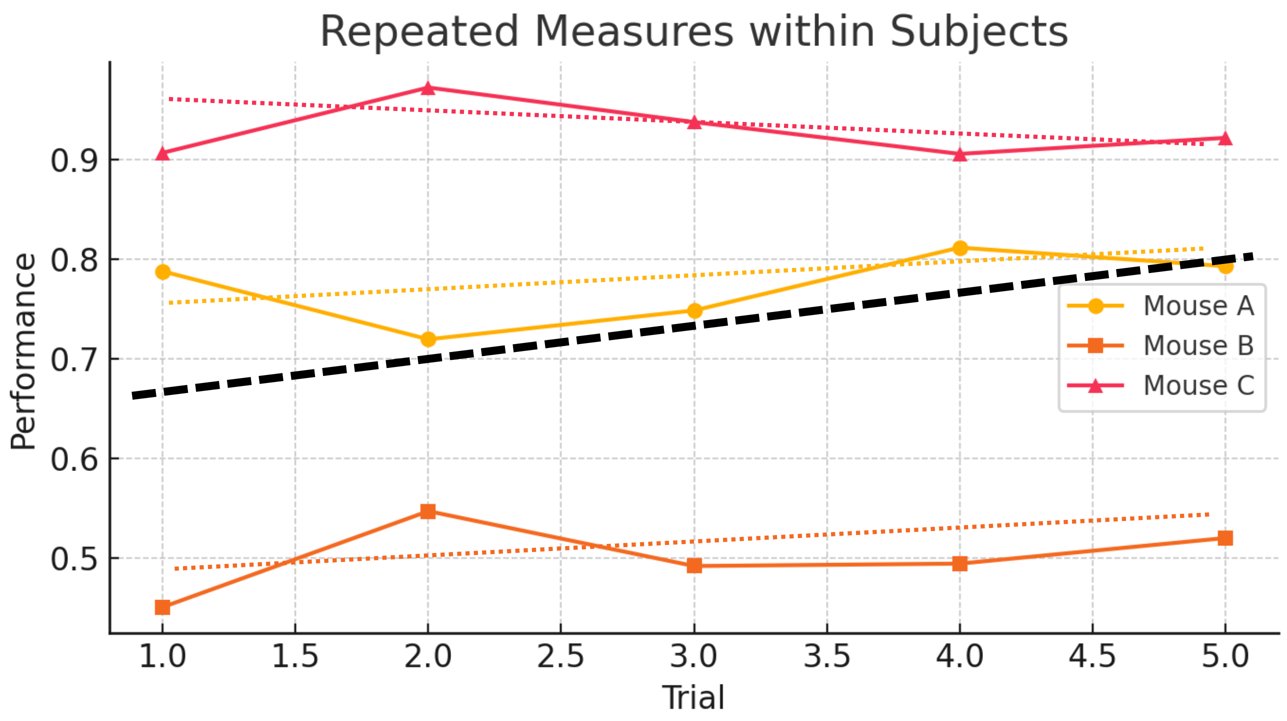


LM Ignores Hierarchical/Nested Structure

Problem: LM flattens the data

Cannot distinguish **within-group** vs **between-group** effects

Example: Mice in Cages
•20 mice from 5 cages
•Mice in the same cage share:

- Diet
- Light cycle
- Handling conditions

LM assumes:
•Each mouse = independent
•Each cage = ignored

Consequences:
•Shared noise gets mistaken for signal
•Generalization fails when moving to **new cages**

Biased Standard Errors in LM

Problem

Linear models **assume all observations are independent**. When this is false (e.g. repeated measures), the model **underestimates uncertainty**.

Consequence

- Standard errors are **too small**
- Confidence intervals are **too narrow**
- p-values become **artificially significant**

More data ≠ more information if the data aren’t independent.

~~**G**eneralised~~

Linear

Mixed-Effects

Models

What is a Linear Mixed-Effects Model?

Motivation

Real-world data is often **grouped, nested, or repeated**: animals in cages, sessions per subject, measurements over time.

LM fails

Key idea

LME models **both fixed effects** (population-level) and **random effects** (group- or subject-level deviations).

LME Model structure

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + u_{0j} + u_{1j} x_{ij} + \varepsilon_{ij}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \rho\tau_0\tau_1 \\ \rho\tau_0\tau_1 & \tau_1^2 \end{pmatrix} \right), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$$

- y_{ij}: outcome for subject j, observation i
- β_0, β_1: fixed intercept and slope
- u_{0j}, u_{1j}: subject-specific random intercept and slope

- τ_0^2: variance of **random intercepts**
- τ_1^2: variance of **random slopes**
- ρ: correlation between intercept and slope
- σ^2: residual (within-subject) variance

All random effects are assumed normally distributed

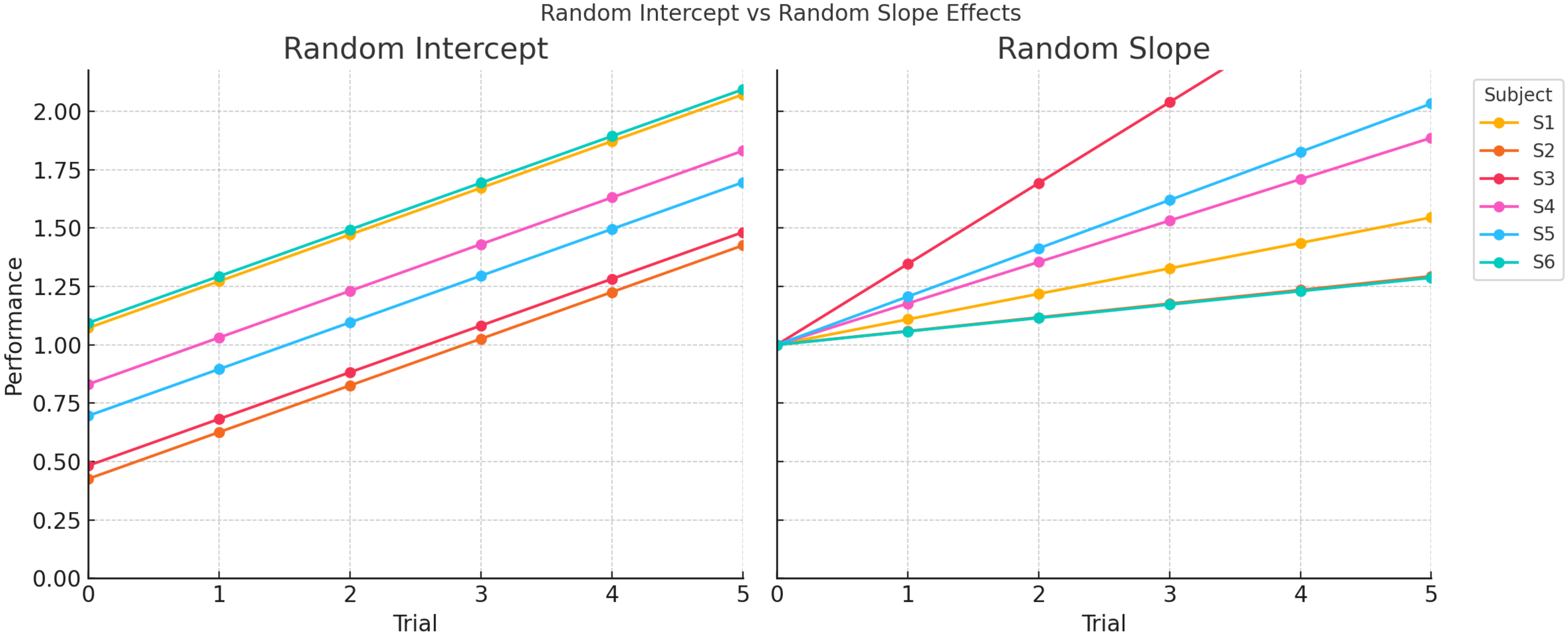
In classical LME, **residual variance σ^2** is assumed to be **constant** across all observations.

Advanced models can relax this assumption, but the default is **homoscedasticity**.

Random Intercept vs Random Slope

Random Intercept Model
All subjects follow the same slope
but each starts from a different baseline (intercept)

Random Slope Model
All subjects share the same intercept
but have different slopes (rates of change)



Parallel lines
Same angle, different vertical positions
Captures **individual level differences in baseline**

Lines start at the same point
Diverge over time or x-axis
Captures **individual trajectories / learning rates**

When to Include Random Intercepts, Slopes, or Both?

Random Intercept
Use when subjects/groups differ in their baseline levels,but follow a similar pattern over predictors.
Example:
- Animals with different baseline activity, but same learning curve
- Imaging sessions from different sites with consistent effect of time

Random Slope
Use when subjects/groups differ in their response to a predictor(i.e. different slopes).
Example:
- Mice improve at different rates across trials
- Treatment effect varies by patient

Random Intercept + Slope
Use when both starting point and rate of change vary across groups.
Example:
- Students in different classrooms with different starting skill and different learning speeds
- Animals with individual reward sensitivity and learning curves

Question	Random effect?
Do subjects differ in average response?	Random **intercept**
Do they react differently to predictors?	Random **slope**
Both?	Use **both**

Experimental Design – Drug Effect on Learning in Mice

Investigate how a drug affects learning over repeated trials in mice.

•**Subjects**: 12 mice, housed in **3 cages** (4 mice per cage)
•**Within-subject manipulation**:
- Each mouse tested in two conditions:
  - **Placebo**
  - **Drug**

•**Task**: repeated trials measuring behavioral **performance** (e.g. accuracy, latency)

Data structure

Variable	Description
`Mouse`	Each mouse observed multiple times
`Cage`	Mice grouped by cage (shared environment)
`Trial`	Repeated trials per condition (within mouse)
`Condition`	Drug vs Placebo (within-subject factor)
`Performance`	Continuous outcome

Key features

Repeated measures: trials × 2 conditions per mouse
Grouped data: mice nested in cages
Within-subject comparison: placebo vs drug
Interested in:
- Learning over trials
- How drug alters learning slope
- Accounting for mouse-level variability
- Controlling for shared cage-level factors

Model Structure – Index Notation + Covariance

We model the **performance** of mouse j from cage k on trial i in condition $c \in \{\text{Placebo}, \text{Drug}\}$ as:

$$y_{ijkc} = \beta_0 + \beta_1 \cdot \text{Trial}_i + \beta_2 \cdot \text{Drug}_c + \beta_3 \cdot (\text{Trial}_i \cdot \text{Drug}_c) + u_{0j} + u_{1j} \cdot \text{Drug}_c + v_k + \varepsilon_{ijkc}$$

Random effects for mouse j

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \rho\tau_0\tau_1 \\ \rho\tau_0\tau_1 & \tau_1^2 \end{pmatrix} \right)$$

- u_{0j}: random intercept (baseline variability across mice)
- u_{1j}: random slope for **Drug** effect
- ρ: correlation between mouse intercept and drug slope

Other random terms

- $v_k \sim \mathcal{N}(0, \omega^2)$: random intercept for cage k
- $\varepsilon_{ijkc} \sim \mathcal{N}(0, \sigma^2)$: residual error

Model in Wilkinson Notation

Formula (Wilkinson syntax)

$$\texttt{Performance} \sim 1 + \texttt{Drug} * \texttt{Trial} + (1 + \texttt{Drug} \mid \texttt{Mouse}) + (1 \mid \texttt{Cage})$$

Wilkinson notation concisely encodes complex model structure:
fixed effects + random effects + interaction terms.

Fixed effects

Term	Meaning
`1`	Global intercept β_0
`Drug`	Main effect of drug β_2
`Trial`	Learning trend β_1
`Drug * Trial`	Interaction: change in learning slope under drug β_3

Random effects

$(1 \mid \texttt{Cage})$ Mice from given cage shared intercept

$(1 + \texttt{Drug} \mid \texttt{Mouse})$ Each mouse has random slope and intercept

Correlated vs Independent Random Effects in lme4

Correlated random intercept and slope (default)

```
model <- lmer(
  Performance ~ Drug * Trial + (1 + Drug | Mouse) + (1 | Cage),
  data = df
)
```

- Mouse-level random effects:
    - Intercept u_{0j}
    - Slope for Drug u_{1j}
- They are modeled with a **2×2 covariance matrix**:

$$\begin{pmatrix} \tau_0^2 & \rho\tau_0\tau_1 \\ \rho\tau_0\tau_1 & \tau_1^2 \end{pmatrix}$$

- ρ is estimated from the data

Independent random effects (diagonal covariance)

```
model <- lmer(
  Performance ~ Drug * Trial + (1 + Drug || Mouse) + (1 | Cage),
  data = df
)
```

- Still includes:
    - Random intercept u_{0j}
    - Random slope u_{1j}
- But they are modeled as **independent**:

$$\mathrm{Cov}(u_{0j}, u_{1j}) = 0$$

- No off-diagonal terms in the covariance matrix

Comparing Models via Likelihood Ratio Test

```r
# Model A: full covariance between intercept and slope
mod_A <- lmer(
  Performance ~ Drug * Trial + (1 + Drug | Mouse) + (1 | Cage),
  data = df, REML = FALSE
)

# Model B: intercept and slope modeled independently
mod_B <- lmer(
  Performance ~ Drug * Trial + (1 + Drug || Mouse) + (1 | Cage),
  data = df, REML = FALSE
)
```

Null hypothesis: intercept and slope for Drug are uncorrelated
Alternative: allow correlation between them improves fit

Likelihood Ratio Test

```r
anova(mod_B, mod_A)
```

- $p < 0.05$ → keep full model: correlation helps
- $p \geq 0.05$ → drop correlation: simpler model sufficient
- This reduces overfitting and improves model stability

Comparing Performance Models Using AIC and BIC

```
AIC(mod_A); BIC(mod_A)
AIC(mod_B); BIC(mod_B)
```

Metric	Penalizes for...	Choose model with...
AIC	model complexity (k)	**lower AIC**
BIC	complexity + n	**lower BIC**

- AIC is more lenient (good for prediction)
- BIC is more conservative (good for inference)

- **ΔAIC > 2** → meaningful difference
- **ΔBIC > 6** → strong evidence for better model
- If both agree: confident choice

Limitations of of Linear Mixed-Effects Models (LME)

Assumption of Gaussian response

•LME requires that the outcome variable is **continuous and normally distributed**

•But many real-world outcomes are:

- **Binary** (correct/incorrect, success/failure)
- **Counts** (number of spikes, errors, visits)
- **Proportions** (0–1, bounded)

Homoscedasticity and symmetric residuals

•LME assumes **constant variance** across fitted values

•Real data often shows:

- Variance increases with the mean
- Skewed distributions (e.g., reaction times, spike counts)

Inappropriate predictions for bounded outcomes

•LME can produce:

- Negative values for counts
- Probabilities outside [0, 1]

•Especially problematic with logistic-like or sparse responses

Takeaway: **LME is powerful**, but it assumes a **Gaussian world**. **GLME extends LME** to handle binary, count, and bounded outcomes using link functions and **non-Gaussian families**.

Generalised
Linear
Mixed-Effects
Models

Limitations of Linear Mixed-Effects Models (LME)

We now allow for **non-Gaussian responses** and link functions:

$$g\left(\mathbb{E}[y_{ij}]\right) = \beta_0 + \beta_1 x_{ij} + u_{0j} + u_{1j} x_{ij}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \rho\tau_0\tau_1 \\ \rho\tau_0\tau_1 & \tau_1^2 \end{pmatrix}\right)$$

- y_{ij}: outcome for subject j, observation i
- β_0, β_1: fixed intercept and slope
- u_{0j}, u_{1j}: subject-specific random intercept and slope
- τ_0^2: variance of **random intercepts**
- τ_1^2: variance of **random slopes**
- ρ: correlation between intercept and slope

•Fixed and random effects structure is **as in LME**
•GLME allows for **non-normal** outcomes and **mean-dependent variance**
•**Link function** transforms linear predictor to match distributional scale

Link Functions in GLME

$$g\left(\mathbb{E}[y_{ij}]\right) = \beta_0 + \beta_1 x_{ij} + u_{0j} + u_{1j} x_{ij}$$

Link functions transform the **mean response** μ_{ij} to the **linear predictor** η_{ij}.

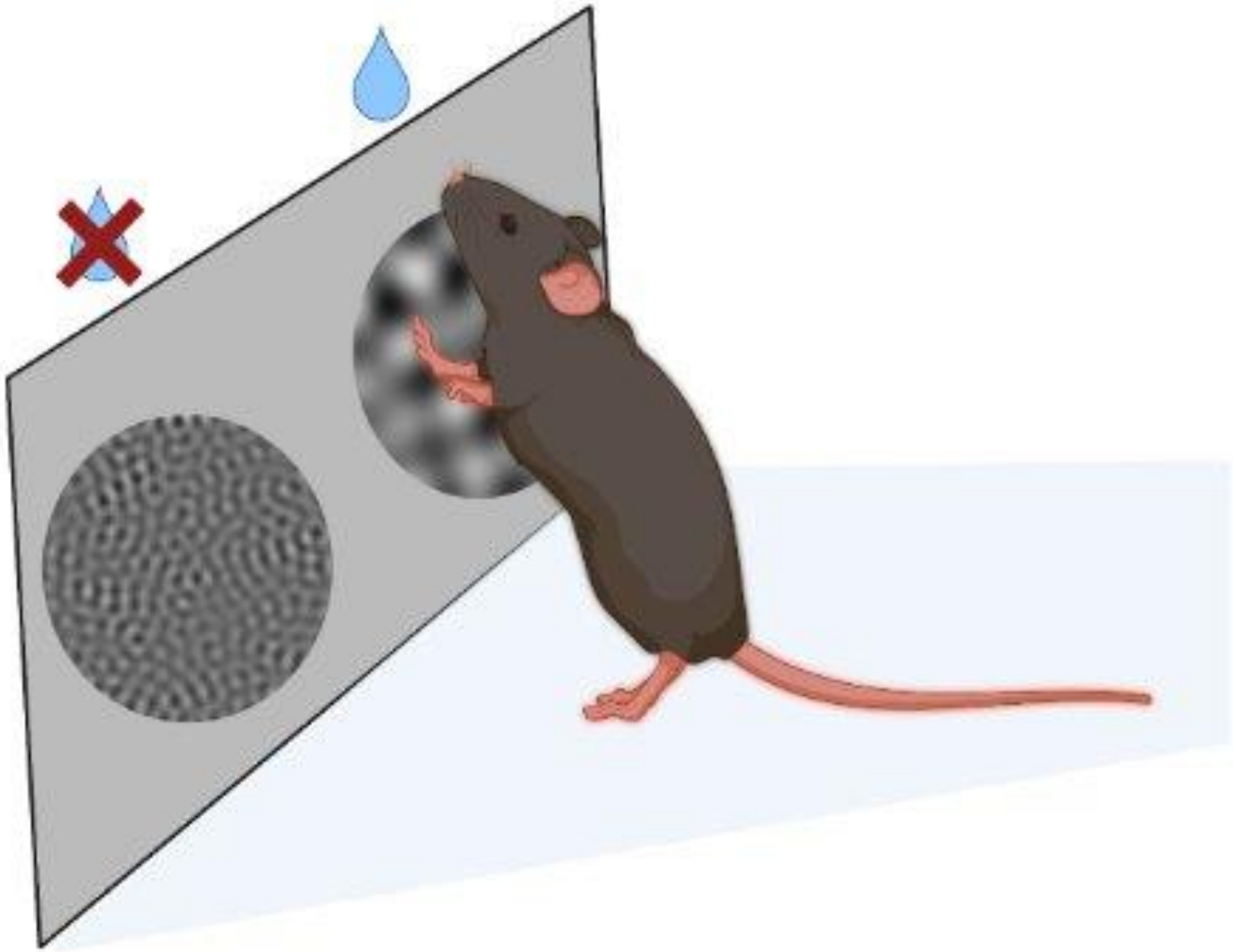
Common link functions:

Response type	Link function $g(\mu)$	Inverse link $g^{-1}(\eta)$
Binomial	$\log\left(\frac{\mu}{1-\mu}\right)$ (logit)	$\mu = \frac{1}{1+e^{-\eta}}$
Poisson	$\log(\mu)$	$\mu = e^{\eta}$
Gaussian (LME)	μ (identity)	$\mu = \eta$

Why link functions matter?

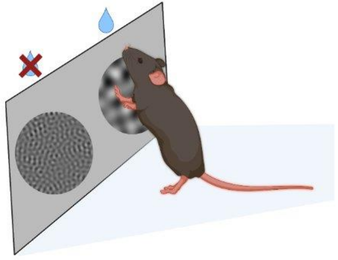
- They control the **shape** of the response space
- Determine **how variance relates to the mean**
- Ensure **valid values** (e.g., no negative counts or >1 probabilities)

Two-Alternative Forced Choice (2AFC) Task in Mice



2-AFC in a mouse touchscreen chamber

Two-Alternative Forced Choice (2AFC) Task in Mice



2-AFC in a mouse touchscreen chamber

Goal: Measure perceptual discrimination and decision-making by presenting **two** visual **stimuli** — **only one is correct**.

Why use GLME?

- **Outcome is binary**: success/failure
- **Multiple trials per mouse** → repeated measures
- Often interested in:
    - Learning over time (Trial)
    - Treatment effects (e.g. Drug)
    - Mouse-level variability (random effects)

Use **logistic GLME** with a **logit link** and **random intercept/slopes per mouse**

FROM EQUATIONS TO EXPERIMENTS